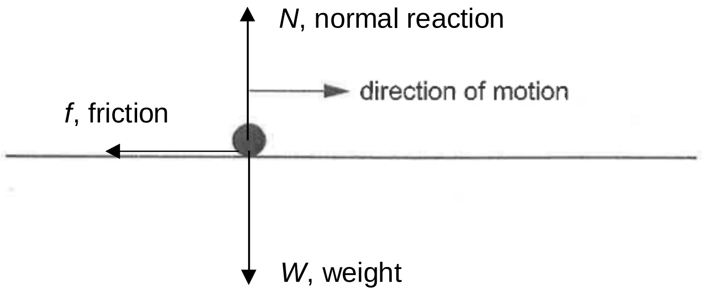
| | |
|---|---|
| | than that of an $\alpha$ particle? |
| **6(a)(i)** | The vertical velocity of the ball decreases at a constant rate as the ball is moving upwards and increases at a constant rate as the ball is moving down. The vertical acceleration of the ball is always downwards and has a constant magnitude of 9.81 m $s^{-2}$. |
| **6(a)(ii)** | As air resistance is negligible, horizontal velocity is constant while the horizontal acceleration is constant at 0 m $s^{-1}$. |
| **6(b)(i)** | Consider vertical motion and taking upwards as positive,<br>$s_y = u_y t + \frac{1}{2} a_y t^2$<br>$0 = (14 \sin 30.0°) t + \frac{1}{2}(-9.81)(t)^2$<br>$t = 1.43 \text{ s}$<br><br>Consider horizontal motion,<br>$s_x = u_x t$ as $a_x = 0$<br>$s_x = (14 \cos 30.0°)(1.43)$<br>$= 17.3 \text{ m}$ |
| **6(b)(ii)** | Consider vertical motion and taking upwards as positive,<br>$s_y = u_y t + \frac{1}{2} a_y t^2$<br>$-68.0 = (14 \sin 30.0°) t + \frac{1}{2}(-9.81)(t)^2$<br>$t = 4.50 \text{ s}$ or $t = -3.08 \text{ s}$ (rejected) |

| No. | Solution |
|---|---|
| | Consider horizontal motion,<br>$s_x = u_x t \quad \text{as } a_x = 0$<br>$s_x = (14\cos 30.0°)(4.50)$<br>$\quad = 54.6 \text{ m}$<br>extra horizontal distance $= 54.6 - 17.3 = 37.3$ m |
| **6(c)(i)** | <br>**Fig. 6.2** |
| **6(c)(ii)** | As the ground is rough, the ball loses K.E. as there is work done against friction as the ball moves along the ground. Once it loses all its K.E., it stops and come to rest. |
| **6(c)(iii)** | With a smaller angle thrown, the horizontal component of its velocity, $u_x = 14\cos 15.0°$ is now larger than its initial horizontal velocity of $14\cos 30.0°$. It will hit the ground with a larger horizontal velocity and will travel a longer distance $L$ before coming to rest. |
| **7(a)** | One similarity is that they have the same number of protons |